Bioinformatics algorithms

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Sequence alignment – exercises

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1. **Local sequence alignment 1**

   Compute the local sequence alignment of the following two strings "DESPERATE" and "ASPARAGUS", using these score parameters:

   - Gap penalty: -2
   - Mismatch penalty: -1
   - Match score: 1

   Answer the following questions:

   (a) What is the final alignment score?

   (b) Are there multiple equivalent solutions? How long are the best local alignments?

   (c) Verify that the final score you calculated is correct by inferring it also directly from the alignment which you obtained (helps to cross-check and identify potential errors in the procedure)!

2. **Local sequence alignment 2**

   Compute the local sequence alignment of the following two DNA sequences "AGCTTAGTTA" and "CGTTATTAG", using these score parameters:

   - Gap penalty: -2
   - Mismatch penalty: -1
   - Match score: 1

   Answer the following questions:

   (a) What is the final alignment score?

   (b) Are there multiple equivalent solutions? How long are the best local alignments?

3. ***Modified* local sequence alignment**

   Given the following two strings: A="AGCTATC" and B="AGCATCTA". How can you find the best local sequence alignment between a *suffix* of string B and any arbitrary substring of A? That is, identify the best possible local alignment which involves some suffix (end) of B.

   Apply your proposed approach using the following scoring parameters:

   - match score = 1
   - mismatch penalty = -1
   - gap penalty = -2

   No need to write any formula, just apply your idea to find the correct solution in the scoring matrix!

4. **Find all local sequence alignments**

   Given the following two strings: A="AATCGTAAT" and B="AACGCAAT", find the following:

   (a) The best local sequence alignment. Is it unique, or are there multiple alignments with this best score?

   (b) Find the second best local sequence alignment.

   (c) Find all local sequence alignments with a minimum score of 2.

5. **Find the best local sequence alignment considering all strands(!)**

   (Note: this exercise is just to make you realize that in biology things are rarely straight forward; you won't need to do such an exercise in the exam ...)

   In practice, if we want to find a proper alignment for two DNA sequences, we would actually need to consider that DNA is double stranded! So the best aligment could also involve not only the input sequences directly, but also the *reverse complementary* strands of one or both input sequences!

   Task: For the two following double-stranded DNA chunks, find the best local sequence alignment considering both of their strands:

   A="TCGTTCAC" and B="CTGAGCGG"